

ZETA-IoT: Four-Layer Zero-Trust Ensemble Architecture

Layer 4: Governance & Compliance

Immutable audit trail • NIST AI RMF mapping • EU AI Act Art. 14/15 controls
Policy versioning • Automated compliance reporting



Layer 3: Tiered AI Governance Engine (TAIGE)

Ensemble detector (IF + OCSVM + Autoencoder) • Window-level decisions
Tiered routing: Autonomous / Escalated / Deferred • Analyst dashboard + feedback



Layer 2: Zero-Trust Microsegmentation

ABAC Policy Decision Point • eBPF/XDP enforcement • Dynamic trust scoring
VLAN/VXLAN segmentation • NGFW integration (optional)



Layer 1: Hardware Root of Trust

TPM 2.0 / ARM TrustZone • Measured boot with PCR sealing
X.509v3 device identity • Hardware-accelerated cryptography

Cross-cutting: key management | logging & monitoring | secure OTA updates | time sync